

ENTERPRISE INFRASTRUCTURE LAB ON MICROSOFT AZURE

End-to-End Deployment of Active Directory, WSUS, File Services, IIS, SQL Server,
Linux Administration, and Backup Solutions

GitHub Repository: <https://github.com/Sreekala86453/Enterprise-Azure-Infrastructure-Lab->

Prepared By
Sreekala Sreedharan

Project Objective

Designed and deployed a multi-tier enterprise infrastructure environment in Microsoft Azure to simulate real-world IT operations. The project demonstrates hands-on experience with identity management, network services, access control, patch management, web hosting, database administration, Linux systems, and backup/recovery operations.

Technologies Used

Microsoft Azure, Windows Server 2022, Active Directory Domain Services, DNS, DHCP, Group Policy, WSUS, SMB File Services, IIS, SQL Server Express, Ubuntu Linux, Apache2, MySQL, Samba, OpenSSH, Windows Server Backup.

Skills Demonstrated

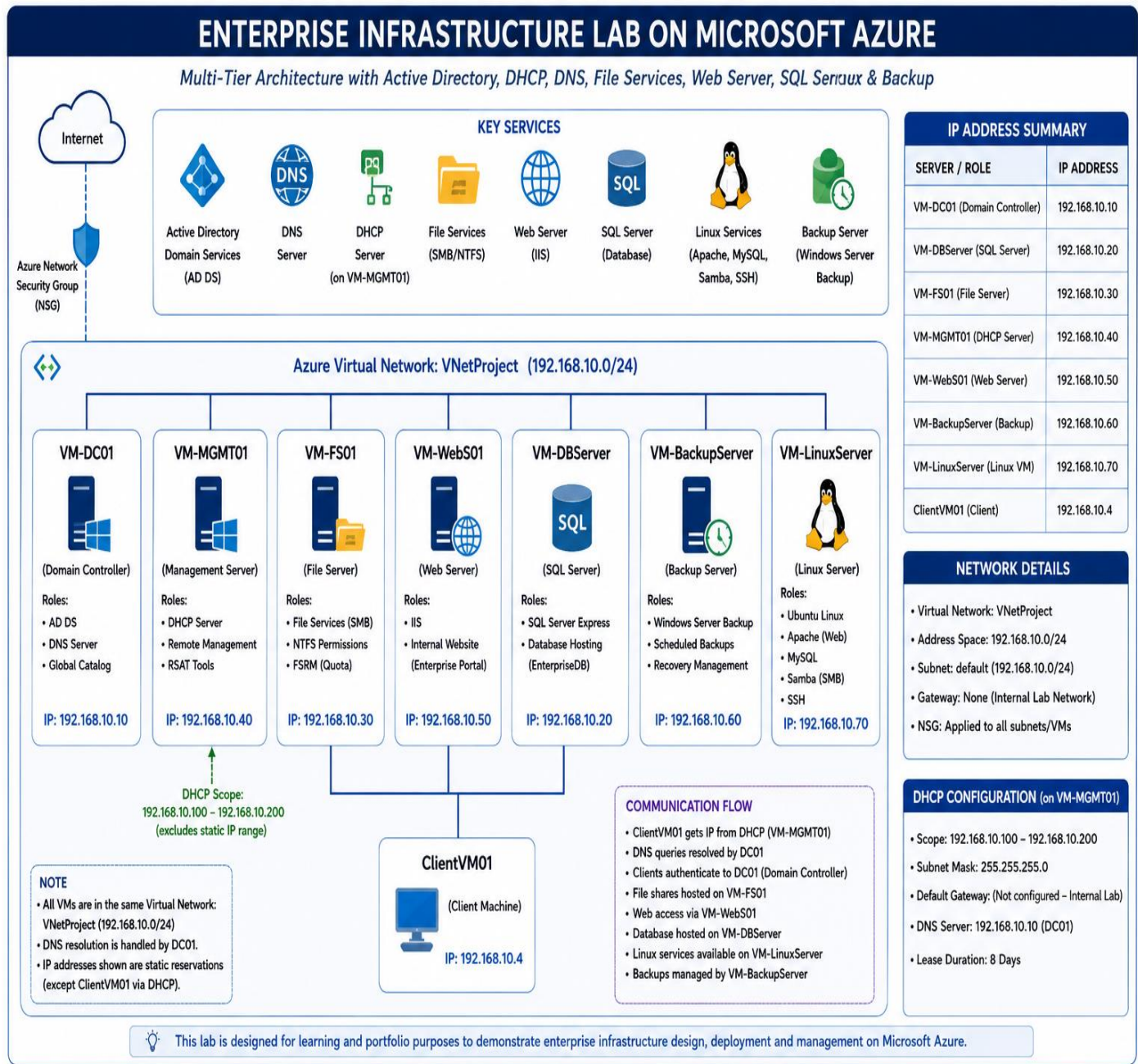
System Administration, Active Directory Administration, Azure Infrastructure, Group Policy Management, Network Services, Security & Access Control, Linux Administration, SQL Administration, Backup & Recovery, and Troubleshooting.

Portfolio Project | Microsoft Azure Enterprise Infrastructure Lab

May 2026-June 2026

Architecture Overview

Designed and deployed a hybrid enterprise infrastructure environment in Microsoft Azure to simulate real-world IT operations, system administration, identity management, patch management, file services, web hosting, SQL services, Linux administration, and backup/recovery processes.



Environment Architecture

Core Infrastructure

- Azure Virtual Network: 192.168.10.0/24
- Active Directory Domain: Enterprise.local
- Resource Group: Project-RG

Server Roles and IP addresses

Server	Role	IP Address
VM-DC01	Active Directory, DNS, Group Policy	192.168.10.10
VM-MGMT01	DHCP + WSUS	192.168.10.40
VM-FS01	File Server +SMB Shares	192.168.10.30
VM-DBServer01	SQL Server Express	192.168.10.20
VM-WebS01	IIS Web Server	192.168.10.50
VM-BackupServer	Windows Server Backup	192.168.10.60
VM-LinuxServer	Ubuntu Linux Server	192.168.10.70
ClientVM01	Windows 11 Domain Client	192.168.10.4

Infrastructure Components

Identity & Directory Services

- VM-DC01 provides centralized authentication and authorization using Active Directory Domain Services.
- DNS services support internal name resolution for all servers and clients.
- Group Policy Objects (GPOs) enforce centralized management and security policies.

Management Services

- VM-MGMT01 provides DHCP services for dynamic IP management.
- WSUS services deliver centralized Windows patch management and update approval.

File Services

- VM-FS01 hosts SMB network shares with NTFS permission-based access control.

- Department-specific shares enforce least privilege access using Active Directory security groups.
- FSRM quotas provide storage governance and capacity management.

Application Services

- VM-WebS01 hosts an internal IIS-based enterprise intranet portal.
- VM-DBServer01 hosts SQL Server Express configured for TCP/IP communication over port 1433.
- Firewall segmentation restricts SQL connectivity exclusively to VM-WebS01.

Linux Services

- VM-LinuxServer provides Linux-based infrastructure services including Apache2, MySQL, Samba, and OpenSSH.
- Cross-platform administration and Linux service management were validated.

Backup & Recovery

- VM-BackupServer performs Windows Server Backup operations for infrastructure systems.
- Backup validation and recovery workflows were tested to simulate business continuity procedures.

Client Validation

- ClieVM01 serves as the domain-joined workstation used for:
 - authentication testing
 - GPO validation
 - file share access testing
 - DNS resolution testing
 - IIS access validation

Technologies Used

Microsoft Infrastructure

- Microsoft Azure
- Windows Server 2022
- Windows 11 Pro
- Active Directory Domain Services (AD DS)
- Group Policy Management
- DNS & DHCP
- WSUS (Windows Server Update Services)
- SMB File Services
- NTFS Permissions
- IIS Web Server
- SQL Server Express
- Windows Server Backup

Linux & Open Source

- Ubuntu Server 22.04
- Apache2
- MySQL Server
- Samba
- OpenSSH

Networking & Security

- Azure Virtual Networks
- Network Security Groups (NSG)
- RDP Restriction by Source IP
- SMB Access Control
- Firewall Rule Configuration
- Role-Based Access Control (RBAC)

Part 1 - Active Directory & Group Policy

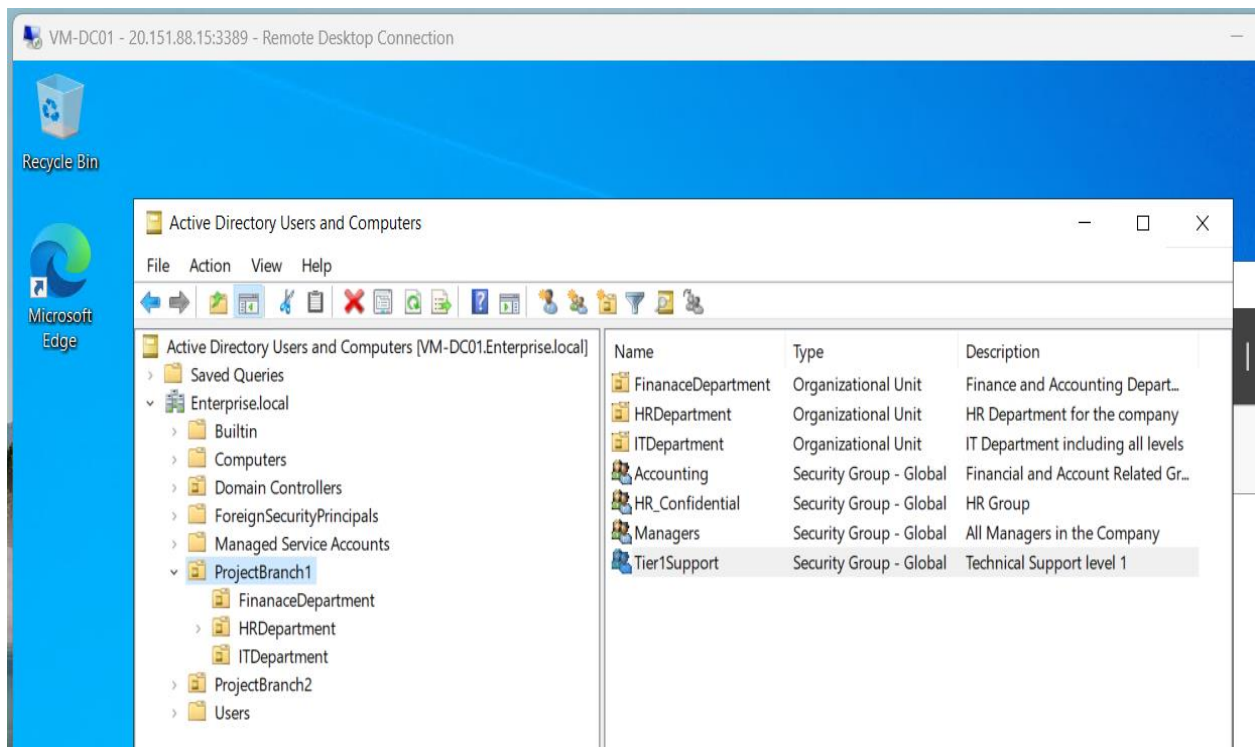
- Deployed Active Directory Domain Services
- Created Organizational Units (OUs)
- Created domain users and security groups
- Implemented role-based access using AD security groups
- Configured Group Policy Objects (GPOs) for Password Policy and Account Lockout

➤ OU Structure and Security Groups:

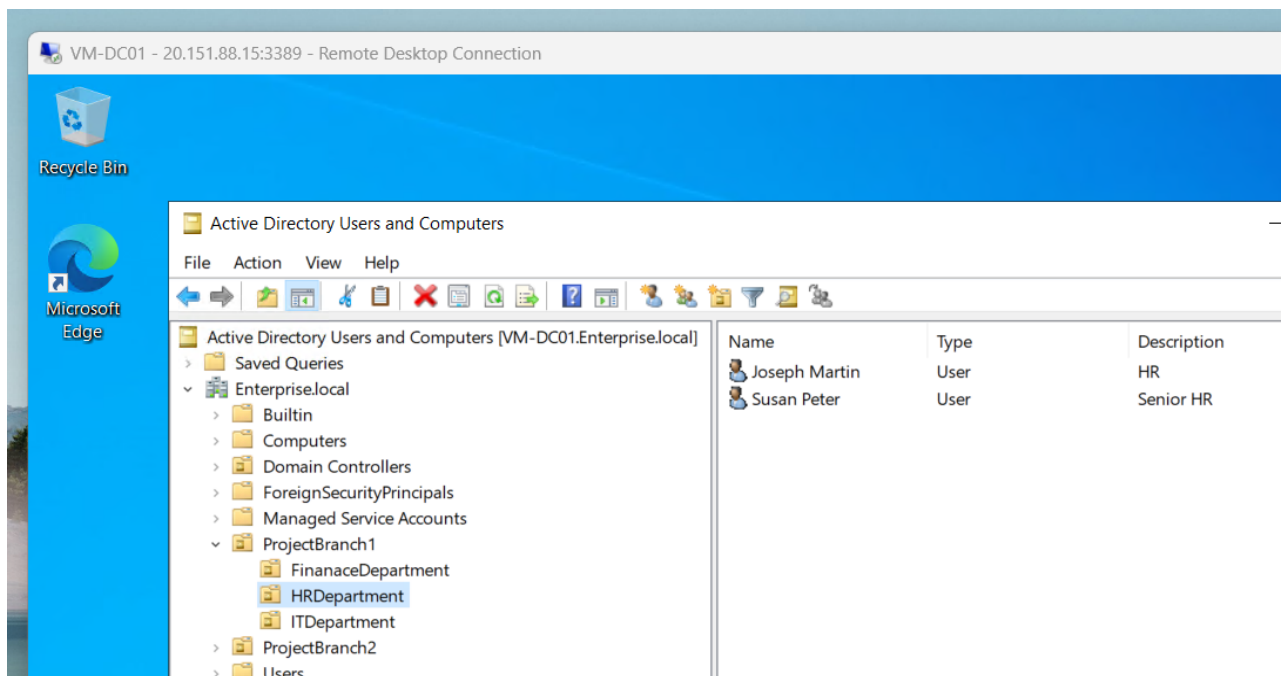
Created 'ProjectBranch1' and 'ProjectBranch2' OUs under the domain 'Enterprise.local'.

Also created Sub-OU's FinanceDepartment, HRDepartment, ITDepartment for allocating users and assigning special permissions in the enterprise level. It is helpful for the management of users and groups more effectively.

Created Global Security Groups in the same OU named 'HR_Confidential', 'Tier1Support', 'Managers' and 'Accounting'.



➤ Users structure

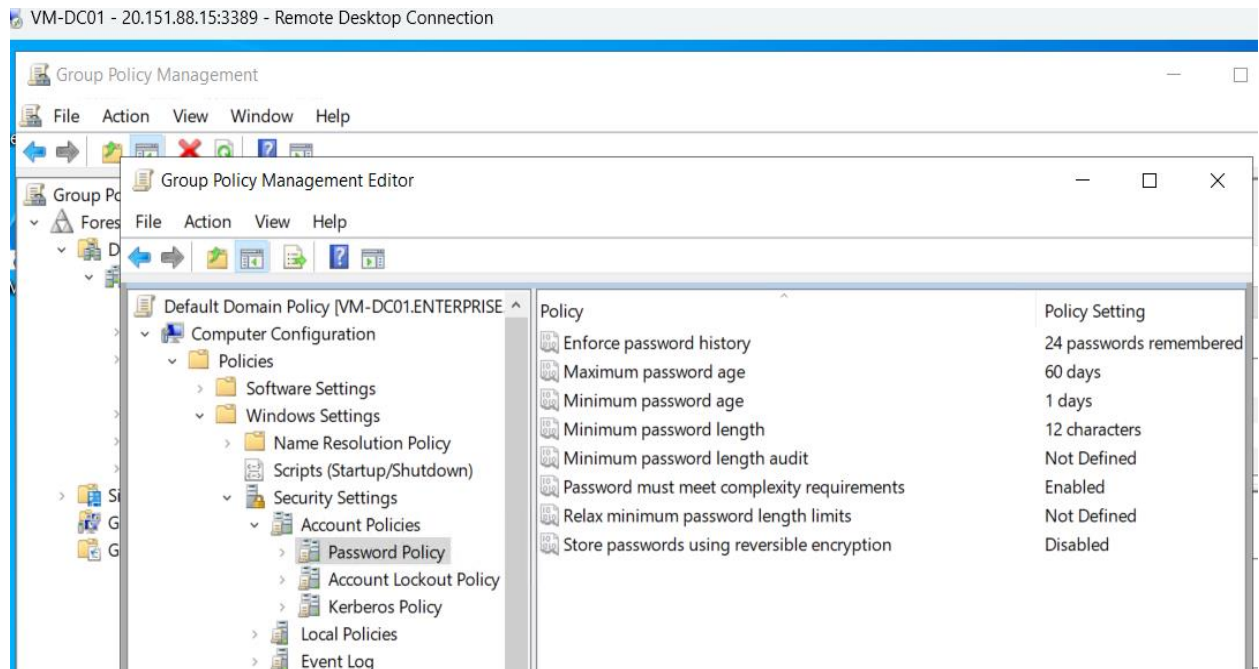


Created 2 users under HRDepartment SubOU, Joseph Martin and Susan Peter.

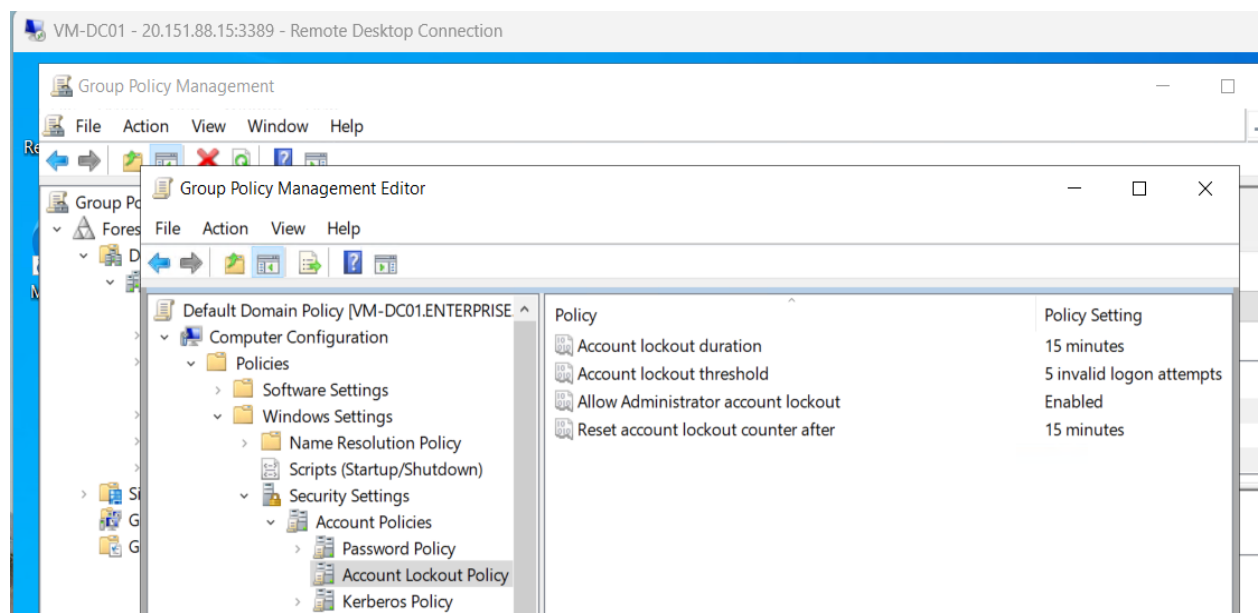
➤ GPO Validation

Created Password Policy GPO and Account Lockout GPO under the default domain.

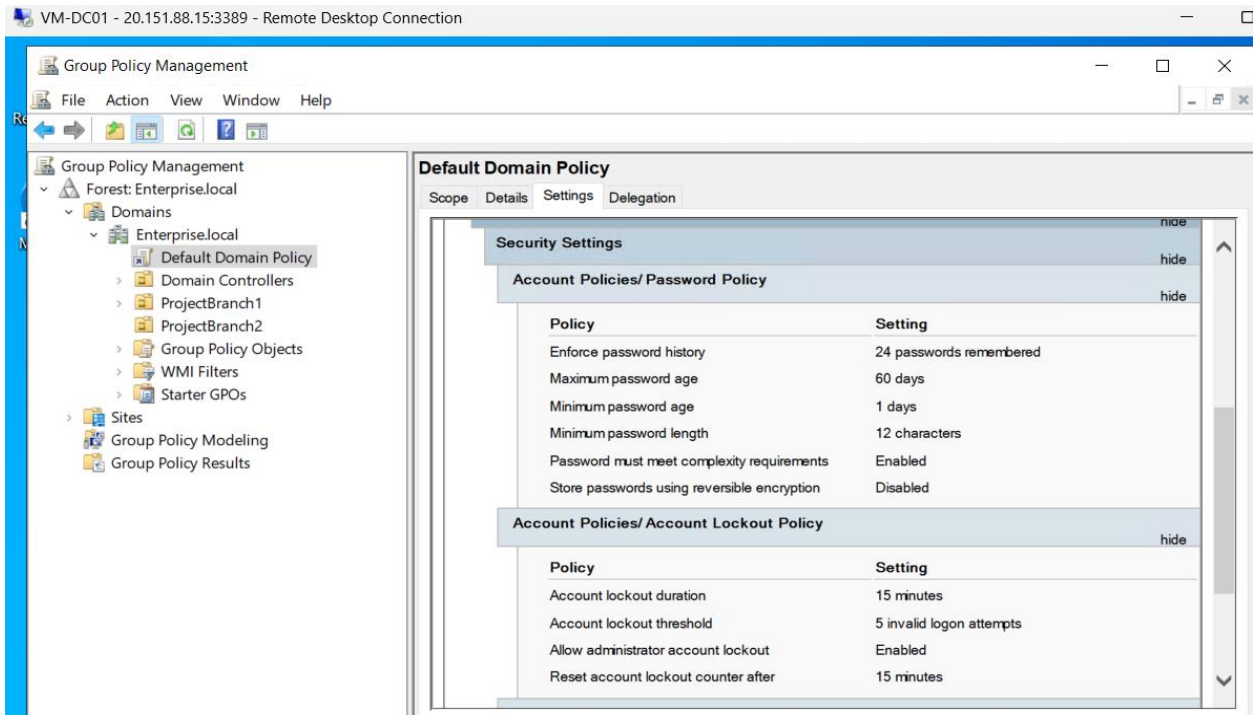
Setting up Password policy GPO on domain controller



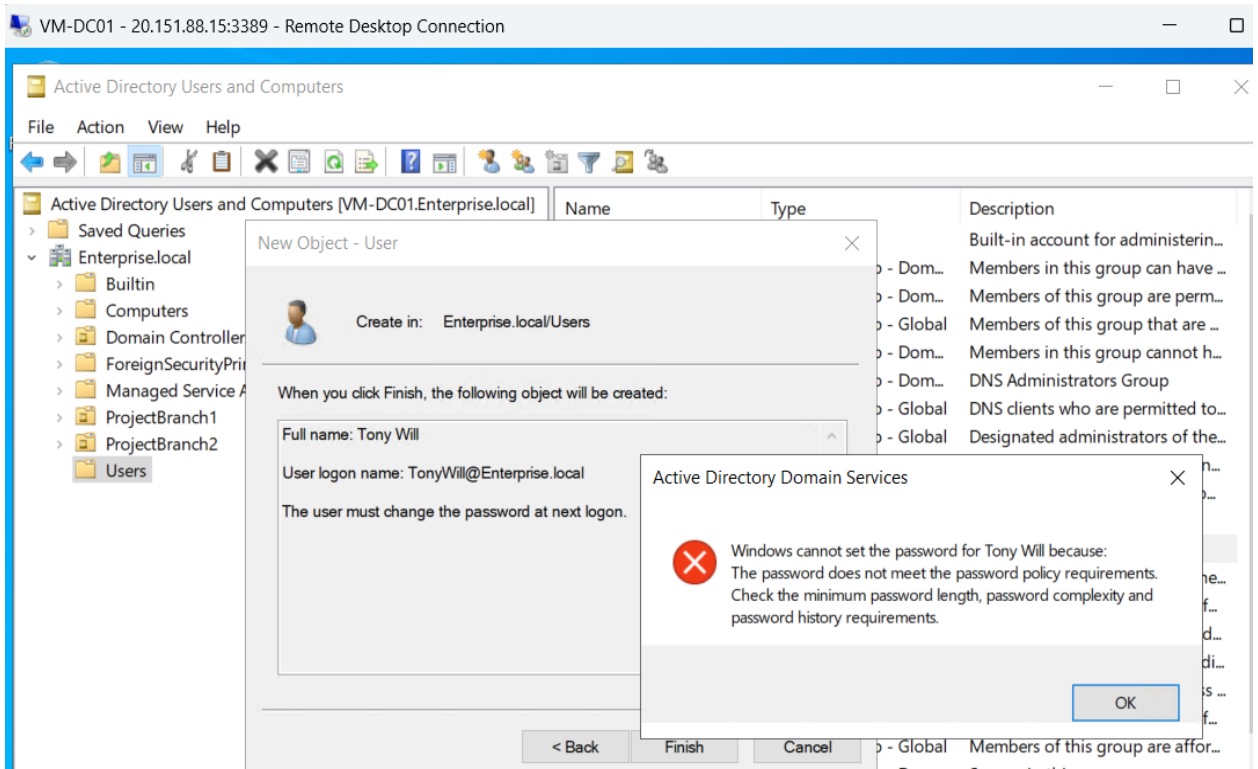
Setting up Account Lockout Policy



Applied Group Policy



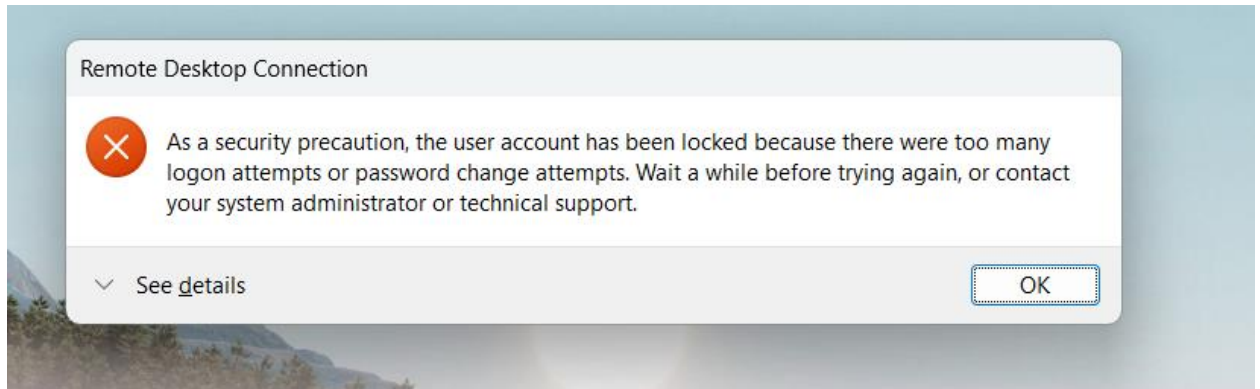
Weak Password Rejection



Tried to create a user 'Tony Will' but password tried was 'Password' which does not meet password complexity created on the group policy, so got error message.

Test result to prove Account Lockout

Tried to login with wrong password for 5 times and on next attempt got the error message.



Part 2 – WSUS & Patch Management

- Deployed WSUS on VM-MGMT01
- Configured update synchronization
- Approved security and critical updates
- Configured clients via Group Policy

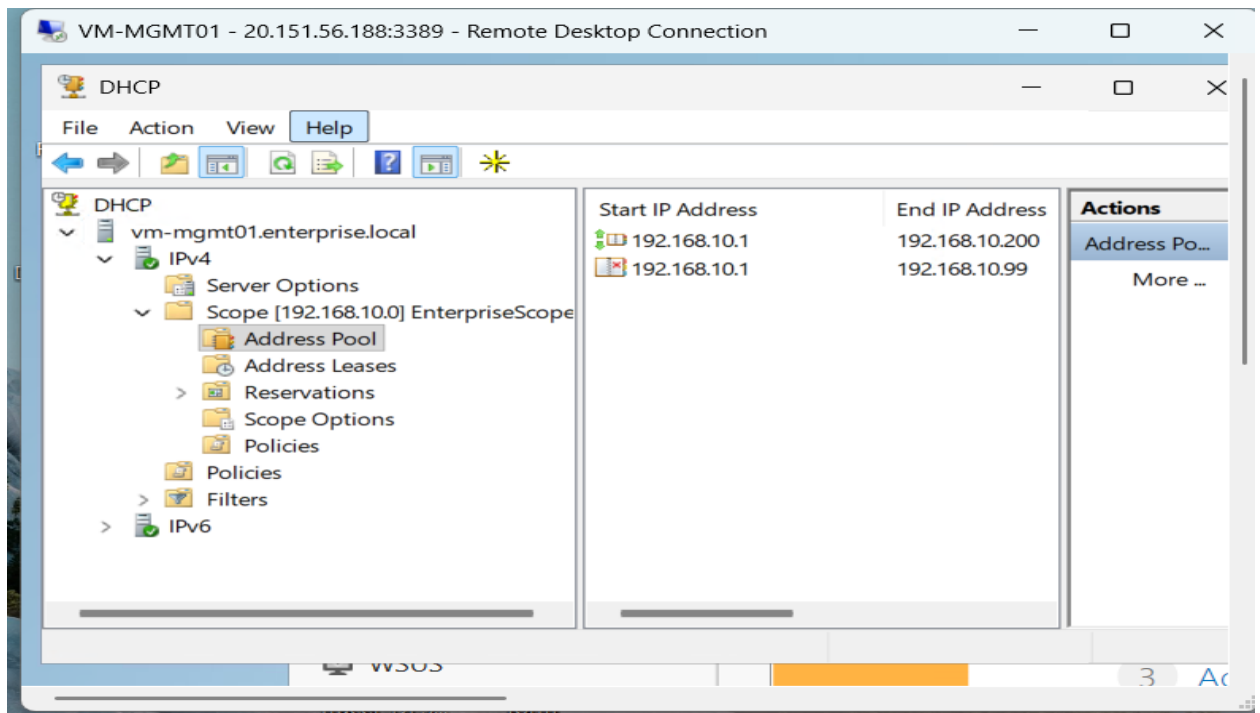
➤ DHCP Configuration

Configured and managed Windows Server DHCP services including IPv4 scopes, reservations, DNS integration, and enterprise network configuration within an Azure-based lab environment.

Here IP Address range: 192.168.10.1 – 192.168.10.200

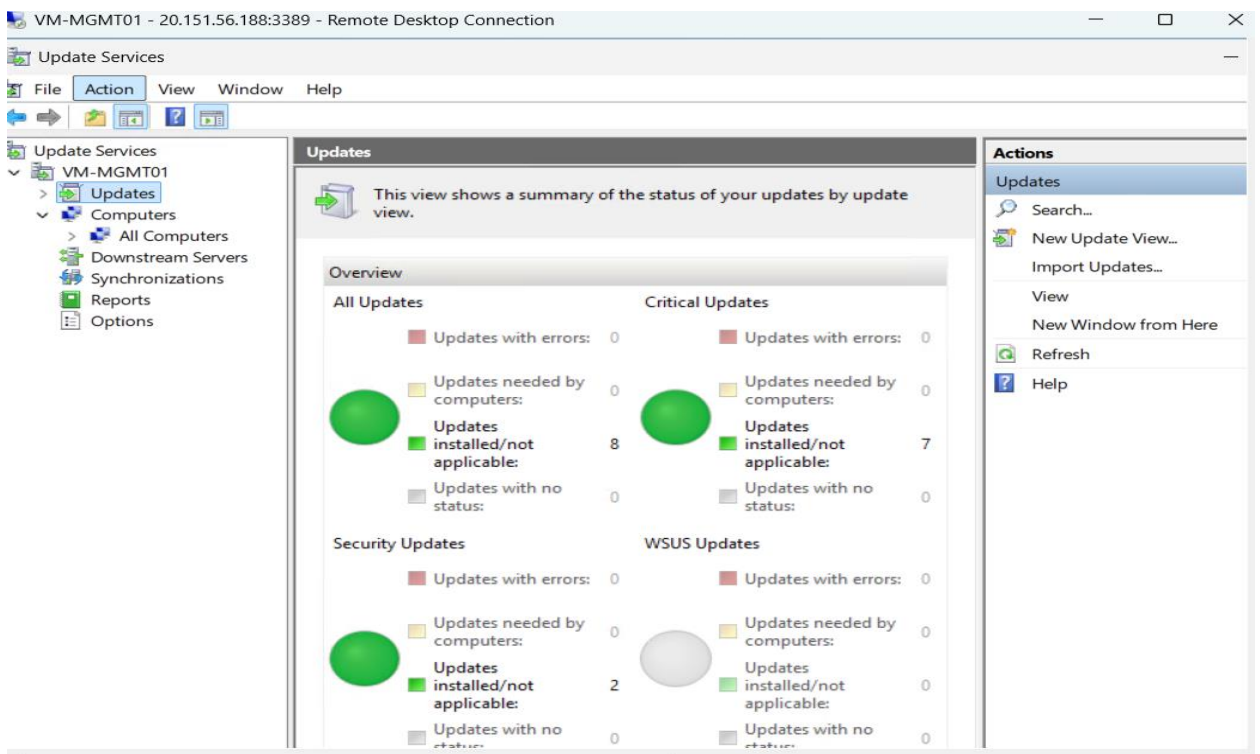
Excluded IP address range: 192.168.10.1- 192.168.10.99 because these range includes lab server environment.

In the Azure environment the machine won't get automatic DHCP IP address from the pool that was created as Azure networking abstracts the real DHCP infrastructure underneath. But this is to show the DHCP Scope creation, DNS integration and enterprise management concepts.



➤ WSUS Dashboard

Implemented centralized patch management using WSUS.



Group policy result for Windows Sever Update Services

```
ClientVM01 - 4.204.40.221:3389 - Remote Desktop Connection

Administrator: Windows PowerShell
PS C:\Windows\system32> gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 5/21/2026 at 3:37:11 AM

RSOP data for ENTERPRISE\AdminS on ClientVM01 : Logging Mode
-----

OS Configuration:                Member Workstation
OS Version:                      10.0.26200
Site Name:                       Default-First-Site-Name
Roaming Profile:                  N/A
Local Profile:                   C:\Users\AdminS
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=ClientVM01,OU=Workstations,DC=Enterprise,DC=local
Last time Group Policy was applied: 5/21/2026 at 3:35:47 AM
Group Policy was applied from:    VM-DC01.Enterprise.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     ENTERPRISE
Domain Type:                     Windows 2008 or later

Applied Group Policy Objects
-----
WSUS-Client-Policy
Default Domain Policy
```

First, I created a folder in the VM-MGMT01 and then installed Windows Server Update services. I selected only security and critical updates during installation and synchronization.

Once it is done, I created a new group policy for WSUS-ClientPolicy on DC01 and forced updated on ClientVM01.

Registry proof of WSUS settings

```
ClientVM01 - 4.204.40.221:3389 - Remote Desktop Connection

Administrator: Windows PowerShell
PS C:\Windows\system32> reg query HKLM\Software\Policies\Microsoft\Windows\WindowsUpdate

HKEY_LOCAL_MACHINE\Software\Policies\Microsoft\Windows\WindowsUpdate
WUserver REG_SZ http://VM-MGMT01:8530
WUstatusServer REG_SZ http://VM-MGMT01:8530
UpdateServiceUrlAlternate REG_SZ
SetProxyBehaviorForUpdateDetection REG_DWORD 0x0

HKEY_LOCAL_MACHINE\Software\Policies\Microsoft\Windows\WindowsUpdate\AU
PS C:\Windows\system32> |
```

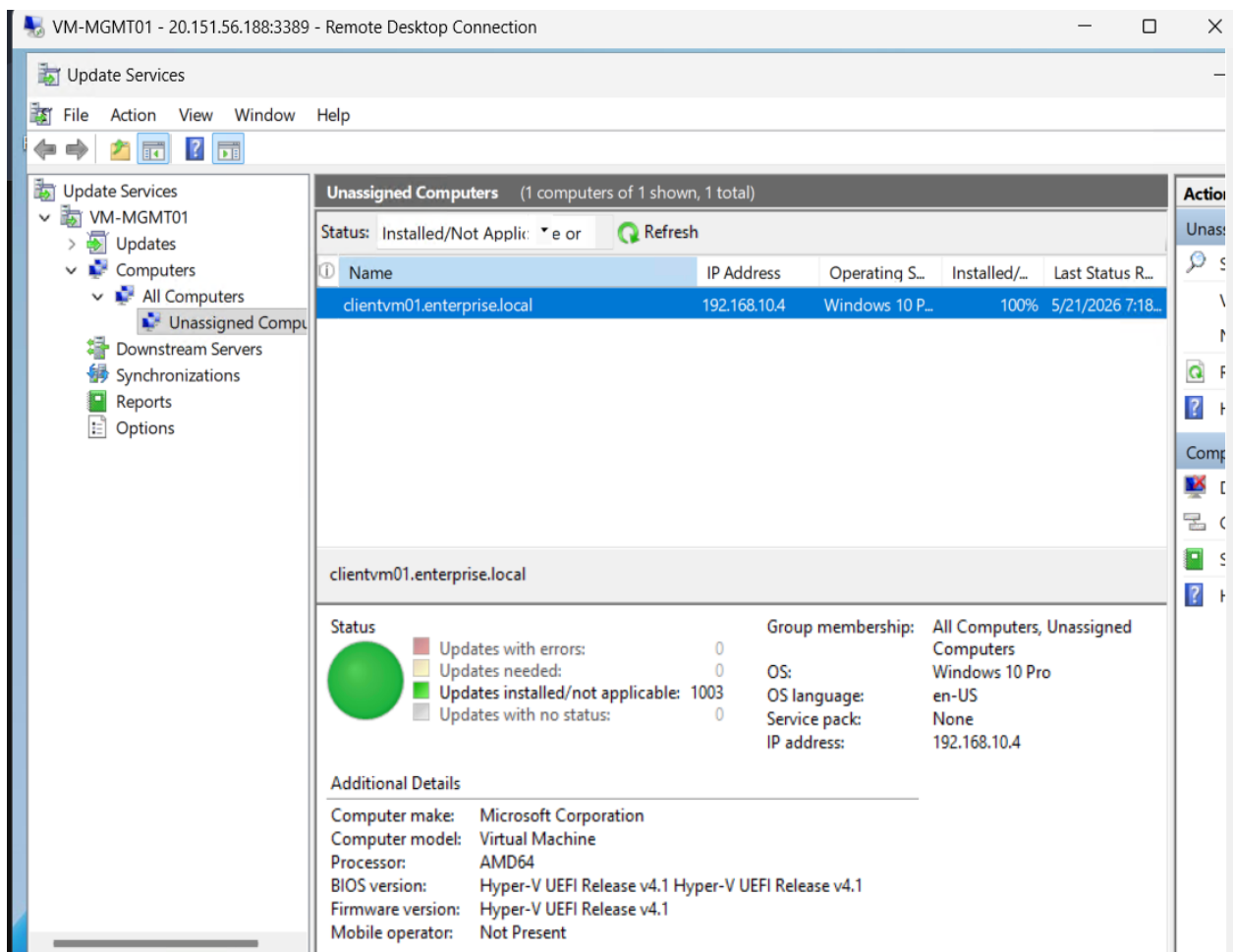
Test Connectivity from ClientVM01 on port 8530

```
ClientVM01 - 4.204.40.221:3389 - Remote Desktop Connection

Administrator: Windows PowerShell
PS C:\Windows\system32> Test-Netconnection VM-MGMT01 -Port 8530

ComputerName      : VM-MGMT01
RemoteAddress     : 192.168.10.40
RemotePort        : 8530
InterfaceAlias    : Ethernet
SourceAddress     : 192.168.10.4
TcpTestSucceeded  : True
```

WSUS Console after triggering the scan

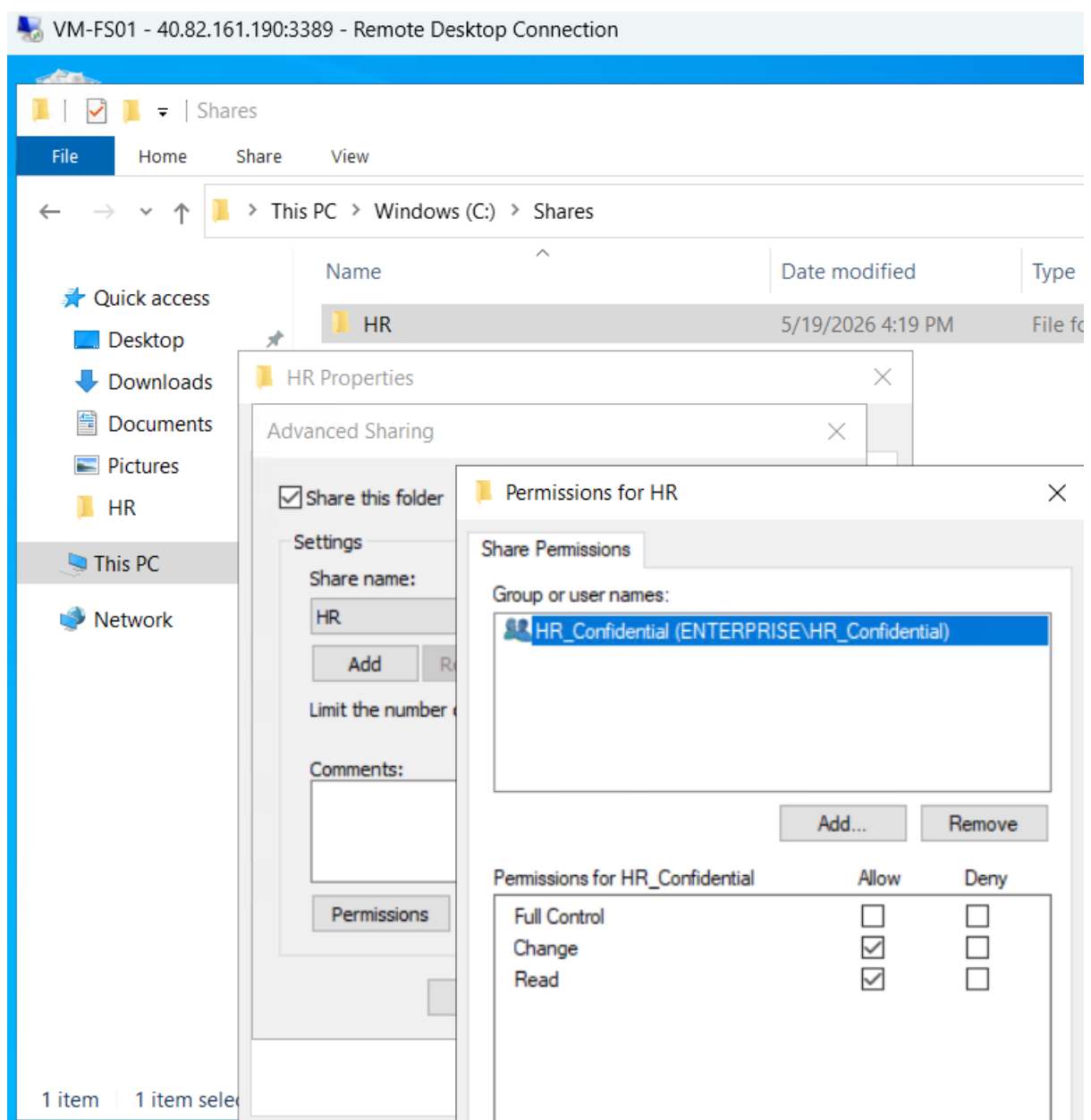


Configured client update settings via Group Policy.

PART 3 – File Server & Access Control

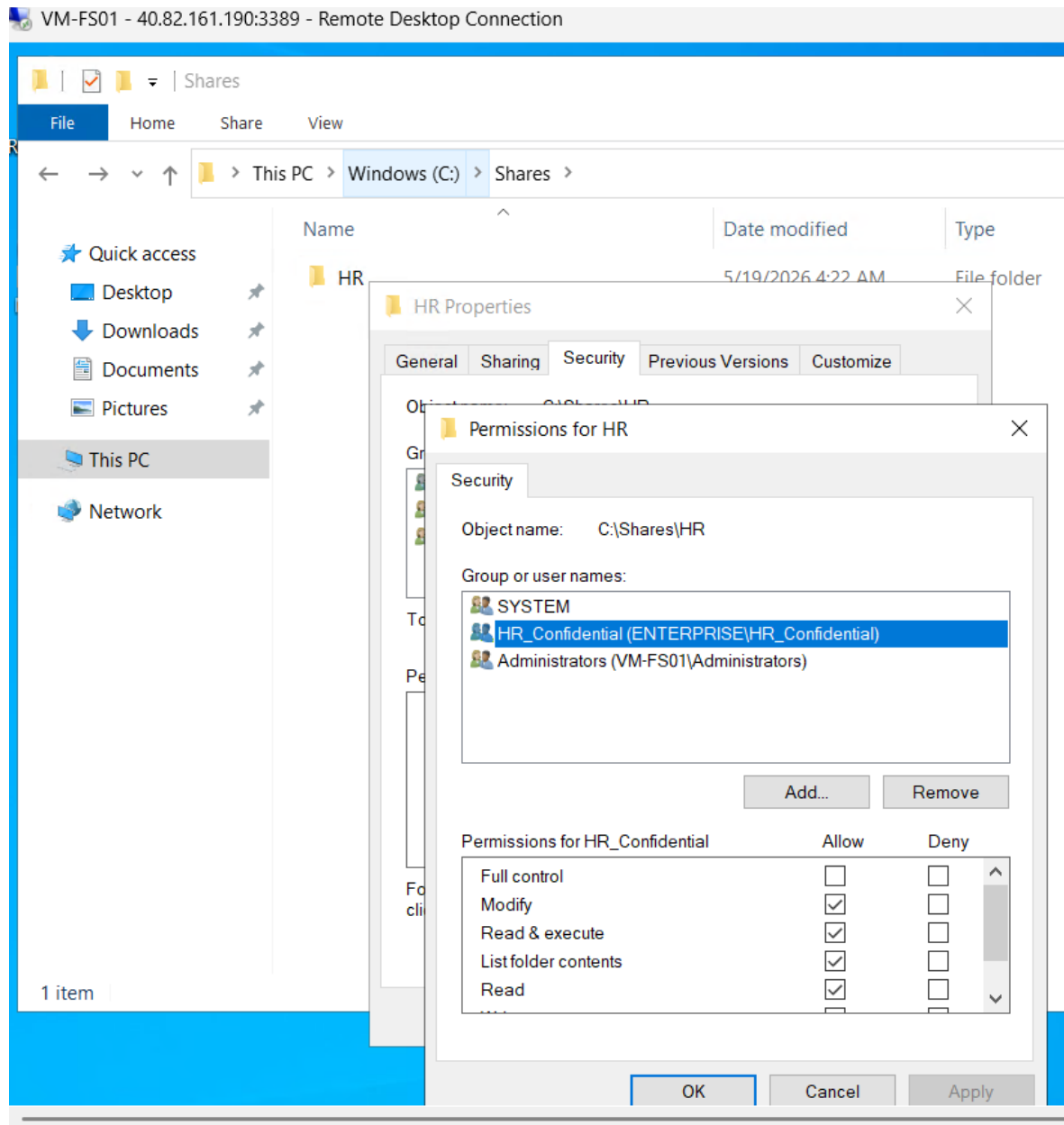
- Created SMB shares:
- \\VM-FS01\HR
- \\VM-FS01\IT
- Configured NTFS permissions on both folders
- Implemented department-based access control
- Validated least privilege access model

➤ Share permission on HR folder



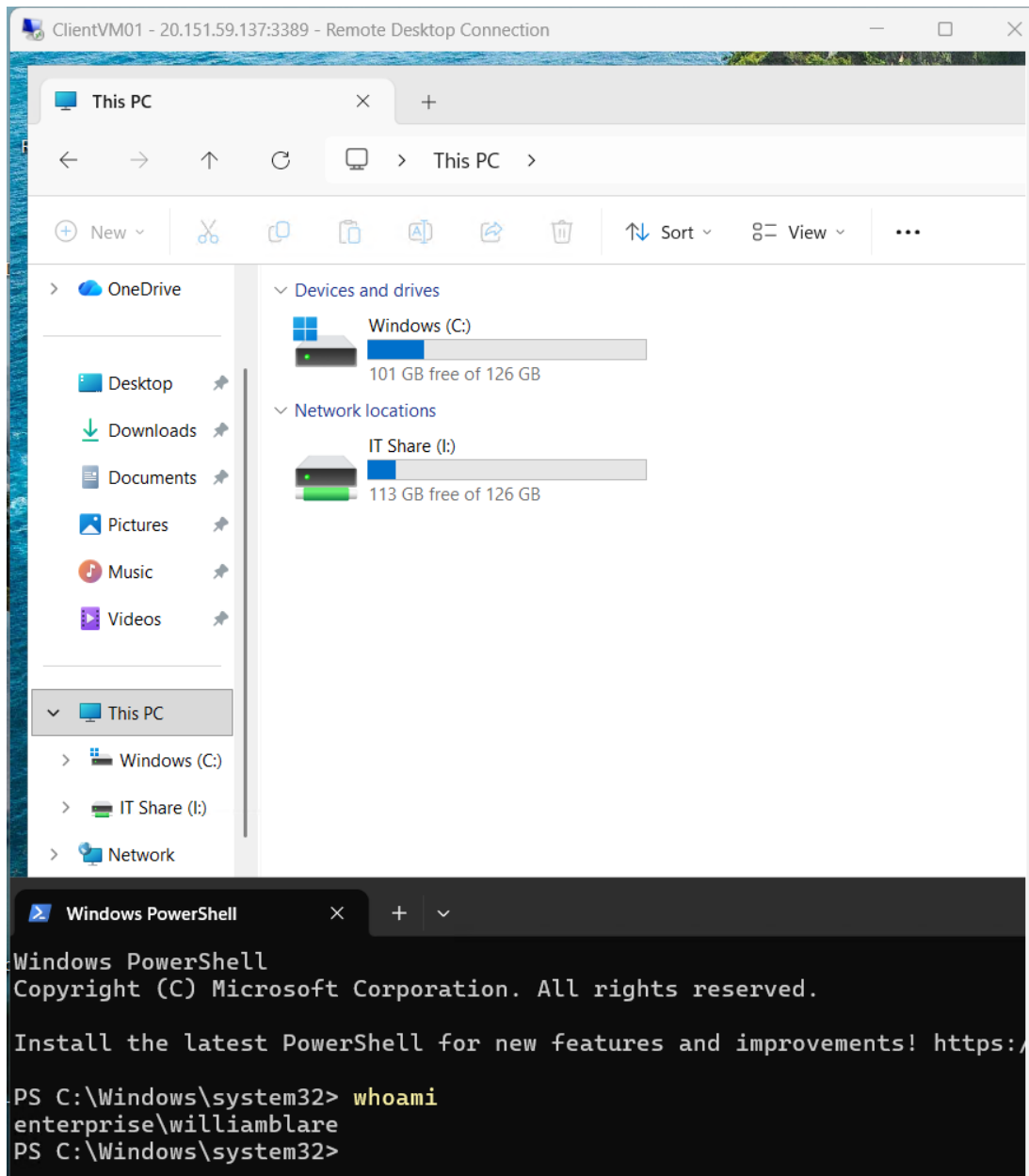
Created two folders HR and IT on VM-FS01. Members of HR_Confidential group only have access to HR Folder. Members can read and change content of the file inside the folder. Also, members of IT_Managers group have access to only IT folder.

➤ NTFS Permission on HR Folder

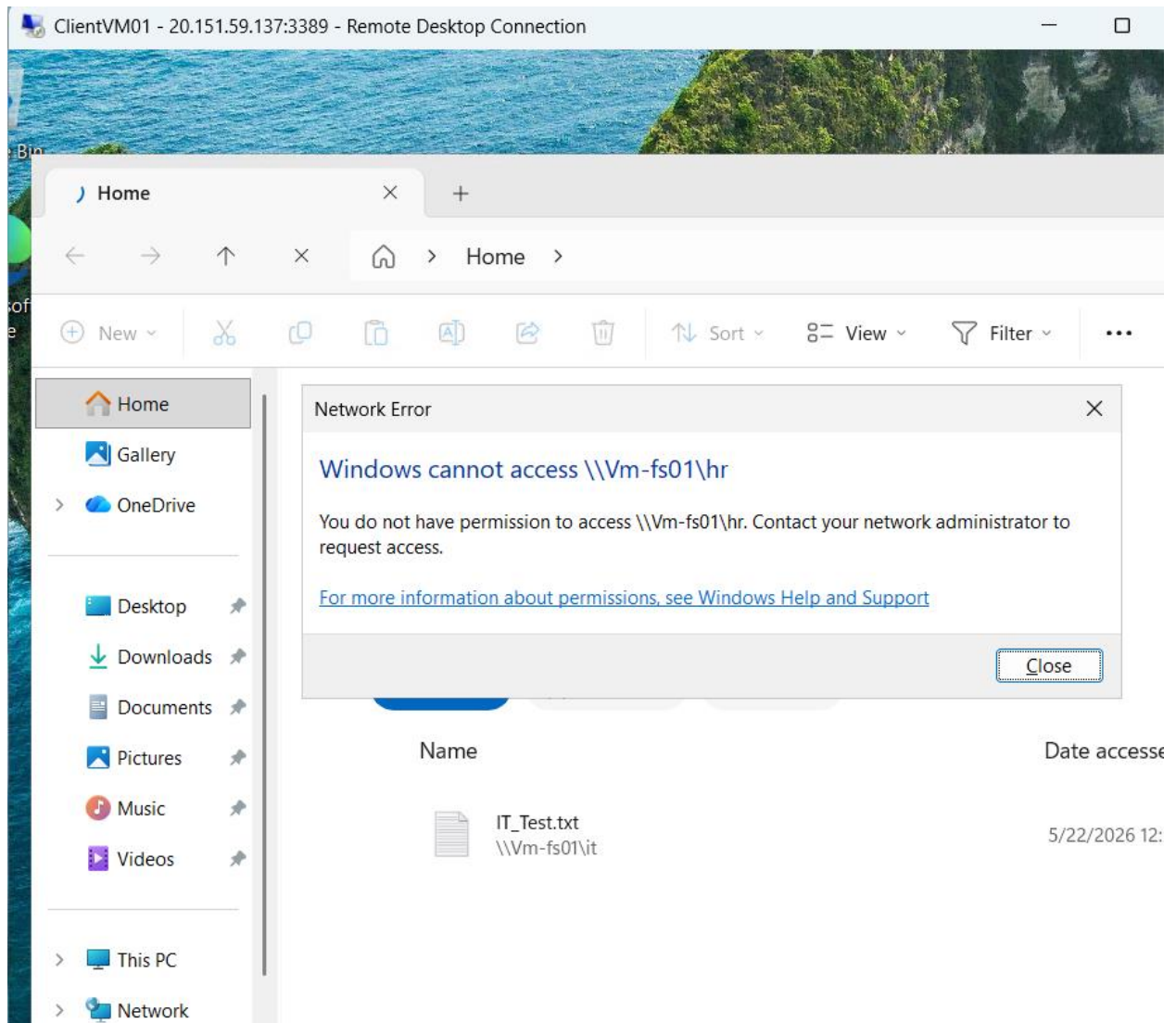


Permission is restricted to HR_Confidential members only.

➤ Access IT Folder by 'William Blare' member of IT_Managers security group.



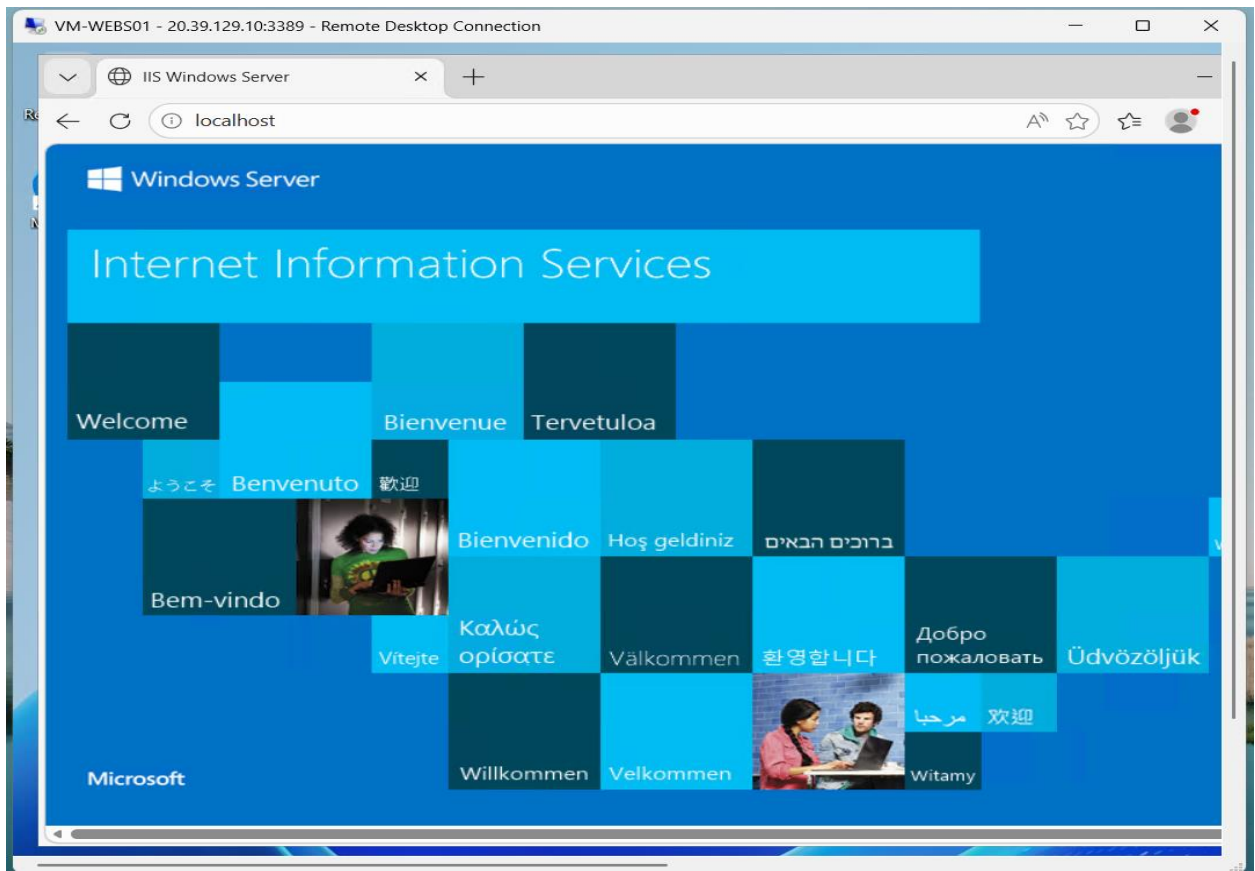
- Access denied when trying to open HR Folder



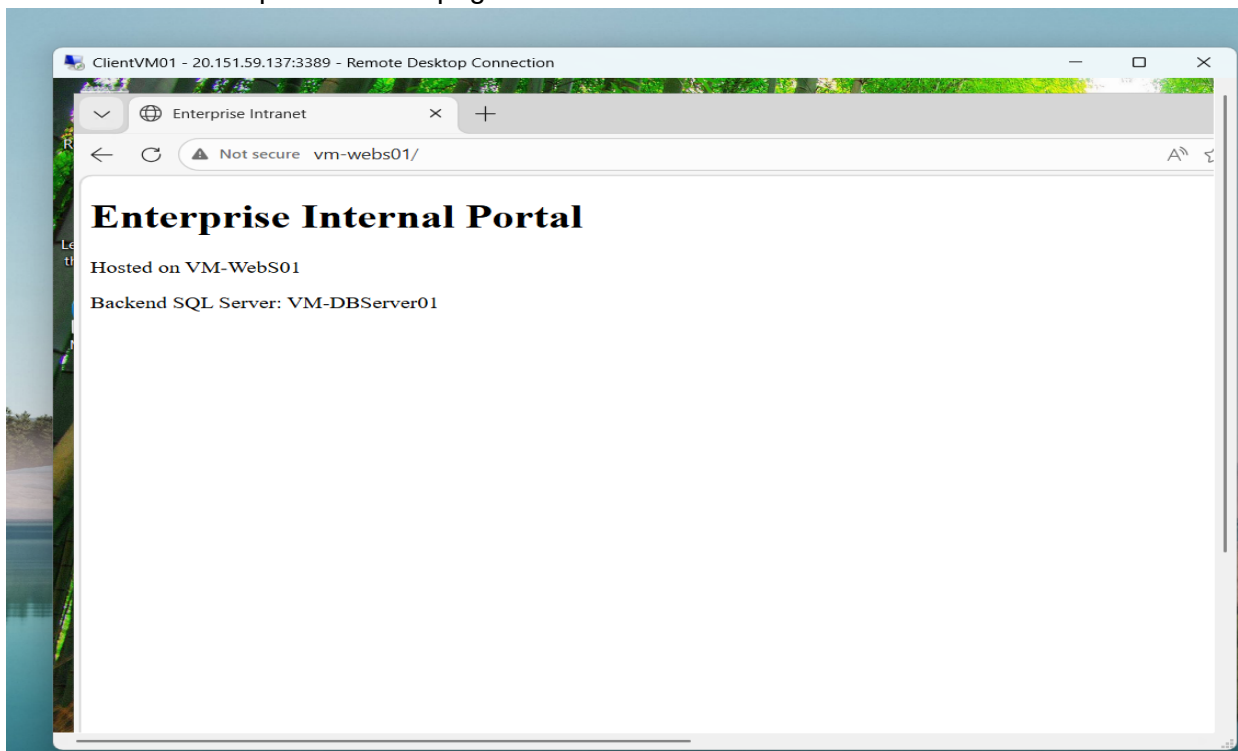
Implemented role-based access control using Active Directory security groups and NTFS permissions.

PART 4 – IIS Server

- Installed and configured IIS on VM-WebS01
- Created internal enterprise intranet page
- IIS Home page screenshot



➤ Access of Enterprise intranet page from web browser

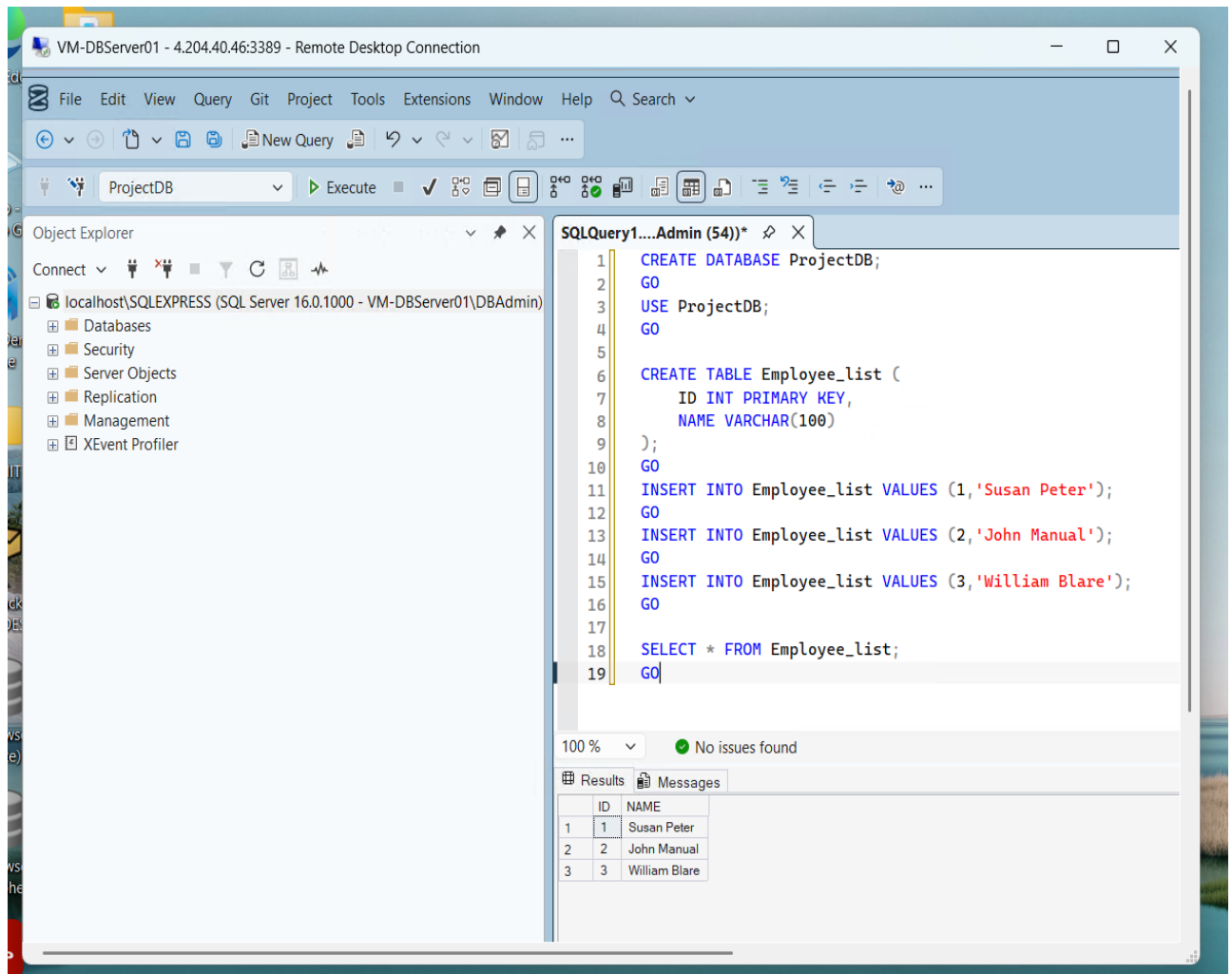


Deployed an internal intranet portal using Microsoft IIS.

PART 5 – SQL Server

- Installed SQL Server Express on VM-DBServer01
- Enabled TCP/IP communication on port 1433
- Configured firewall rules to restrict SQL access to VM-WebS01 only
- Created and validated SQL database objects using SSMS

➤ SQL Validation



PART 6 – Linux Administration

- Deployed Ubuntu Server in Azure
- Installed Apache2, MySQL, Samba, and OpenSSH
- Validated SSH remote administration
- Validated Apache web services
- Configured Linux service management

➤ SSH Login on VM-LinuxServer

```
LinuxAdmin@VM-LinuxServer X + v
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\sreekala> ssh LinuxAdmin@40.82.170.236
LinuxAdmin@40.82.170.236's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1015-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat May 23 01:39:41 UTC 2026

System load:  0.0           Temperature:   49.9 C
Usage of /:   8.9% of 28.02GB Processes:    138
Memory usage: 18%          Users logged in: 1
Swap usage:   0%           IPv4 address for eth0: 192.168.10.70

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection

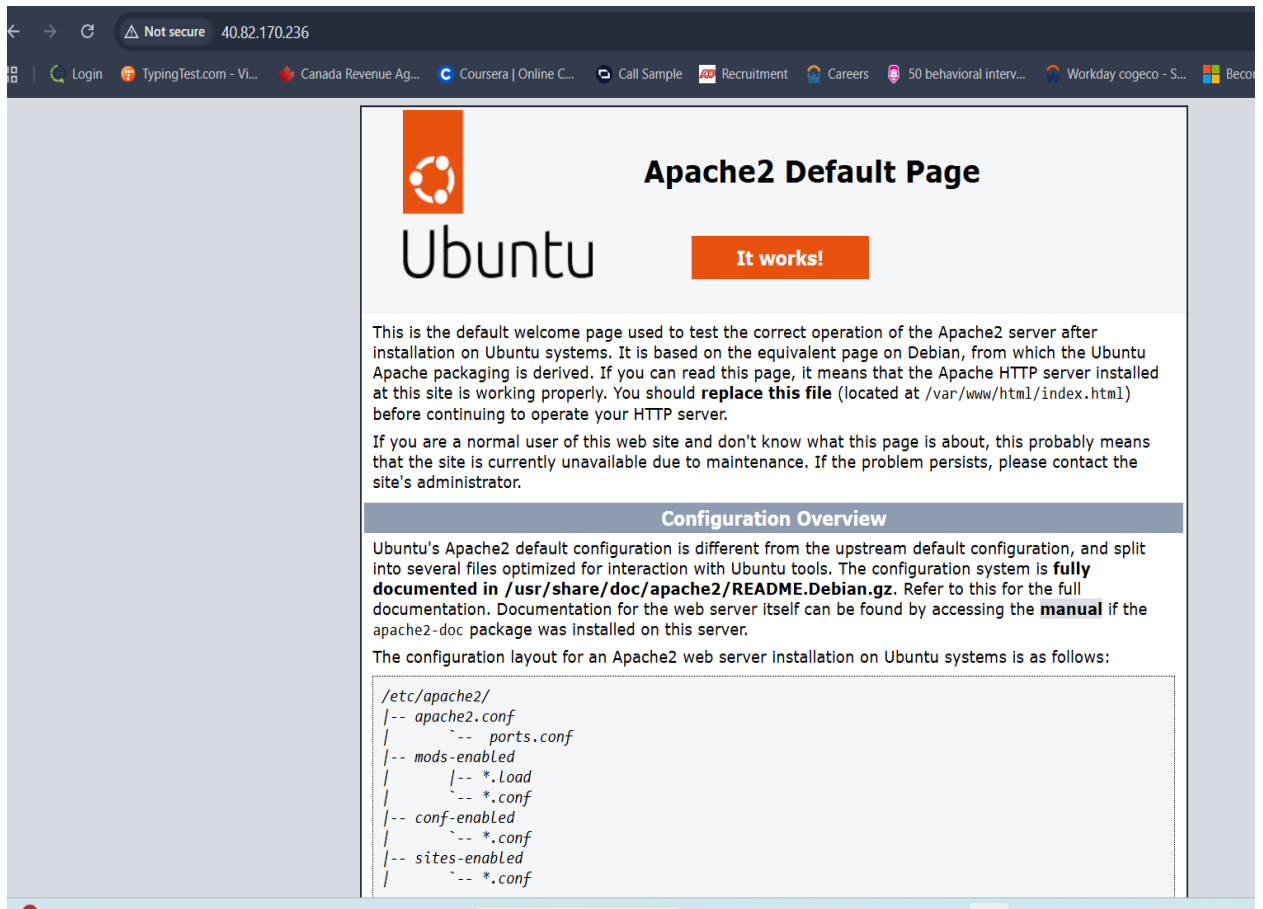
Last login: Sat May 23 01:13:22 2026 from 173.32.79.36
LinuxAdmin@VM-LinuxServer01:~$
```

➤ Status of Apache on Linux VM

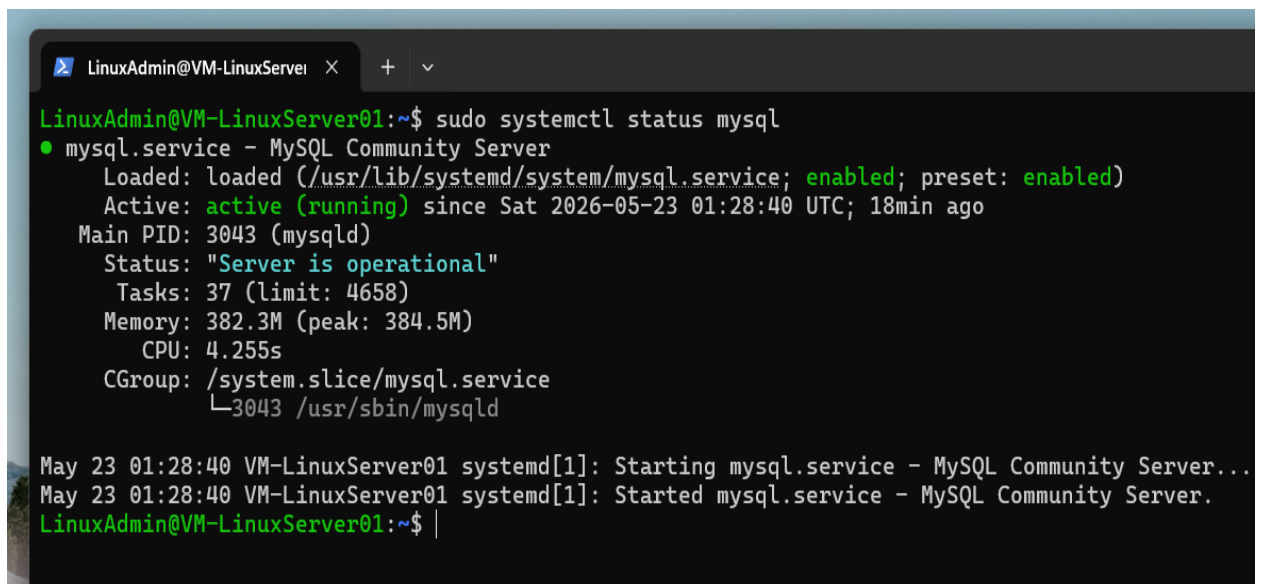
```
LinuxAdmin@VM-LinuxServer X + v
LinuxAdmin@VM-LinuxServer01:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-23 01:28:46 UTC; 3min 3s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3477 (apache2)
    Tasks: 55 (limit: 4658)
   Memory: 5.2M (peak: 5.7M)
      CPU: 34ms
   CGroup: /system.slice/apache2.service
           └─3477 /usr/sbin/apache2 -k start
             └─3479 /usr/sbin/apache2 -k start
               └─3480 /usr/sbin/apache2 -k start

May 23 01:28:46 VM-LinuxServer01 systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 23 01:28:46 VM-LinuxServer01 systemd[1]: Started apache2.service - The Apache HTTP Server.
LinuxAdmin@VM-LinuxServer01:~$
```

➤ Web page Access from local browser



➤ Status of MySQL



➤ Status of Samba Service

```
LinuxAdmin@VM-LinuxServer01:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-23 01:28:48 UTC; 20min ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
    Main PID: 3915 (smbd)
      Status: "smbd: ready to serve connections..."
        Tasks: 3 (limit: 4658)
      Memory: 10.4M (peak: 11.1M)
         CPU: 51ms
    CGroup: /system.slice/smbd.service
            └─3915 /usr/sbin/smbd --foreground --no-process-group
              └─3918 "smbd: notifyd" .
                └─3919 "smbd: cleanupd "
```

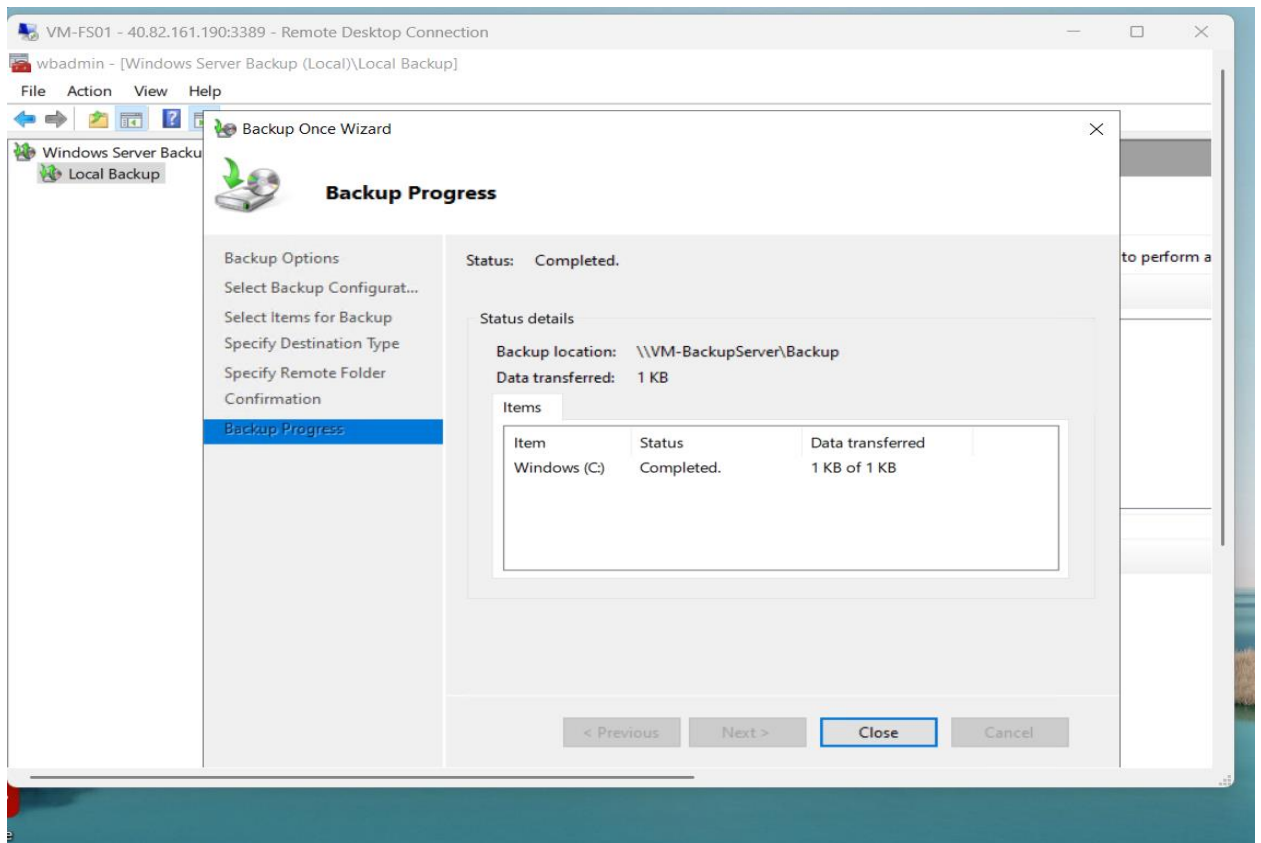
```
May 23 01:28:48 VM-LinuxServer01 systemd[1]: Starting smbd.service - Samba SMB Daemon...
May 23 01:28:48 VM-LinuxServer01 (smbd)[3915]: smbd.service: Referenced but unset environment va
May 23 01:28:48 VM-LinuxServer01 systemd[1]: Started smbd.service - Samba SMB Daemon.

LinuxAdmin@VM-LinuxServer01:~$
```

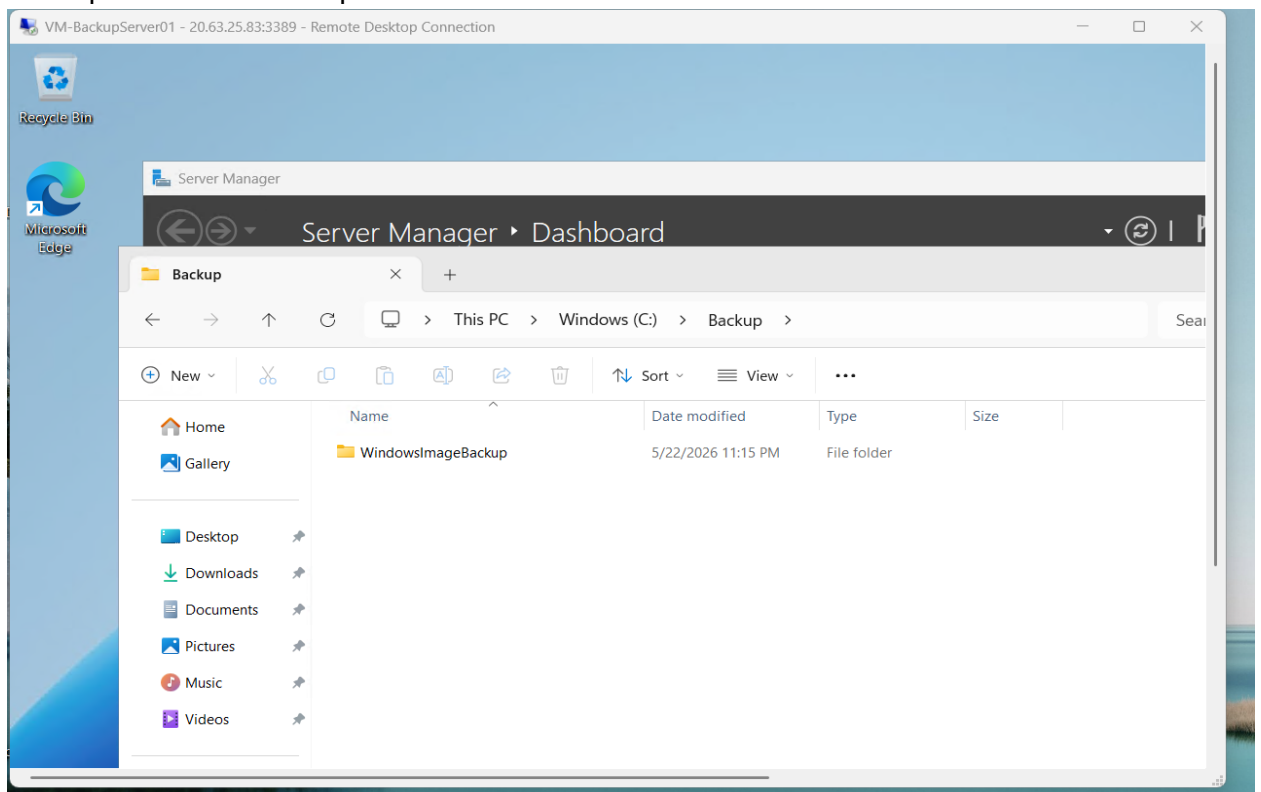
Configured Apache, MySQL, Samba, and OpenSSH services on Ubuntu Server.

PART 7 – Backup & Recovery

- Installed Windows Server Backup
 - Configured backup operations for infrastructure systems
 - Validated successful backup completion
 - Tested recovery workflow concepts
- Back up Services completed successfully on VM-FS01



➤ Backup File on the Backup Server



Challenges Encountered

- Encountered and troubleshooted Group Policy application failures.
- Corrected inherited permissions to enforce least privilege access.
- Validated WSUS communication on TCP 8530.

Validation & Testing Performed

Active Directory Validation

- Successful domain authentication
- Domain user login testing
- Group membership validation

DNS & Network Validation

- Verified DNS name resolution using nslookup
- Verified server communication using ping and Test-NetConnection
- Validated SQL connectivity on TCP 1433

File Share Validation

- HR users successfully accessed HR shares
- Access denied validation for unauthorized shares
- Verified drive mapping GPO deployment

Web & SQL Validation

- IIS intranet page successfully hosted and accessed
- SQL database connectivity validated
- SQL queries executed successfully through SSMS

Linux Validation

- SSH connectivity validated
- Apache web page successfully accessible
- Linux services verified through systemctl

Backup Validation

- Successful Windows Server Backup job completion
- Backup status and logs verified

Key Skills Demonstrated

Area	Technologies
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Infrastructure Administration	Active Directory, DNS, DHCP, Group Policy
System Administration	Windows Server 2022, Windows 11, Ubuntu Linux
Security & Access Control	RBAC, NTFS Permissions, SMB shares, Network Security Groups
Application Services	IIS, SQL Server Express
Operations	WSUS, Windows Server Backup, Service Monitoring, Troubleshooting
Database	SQL Server Express
Linux	Ubuntu, Apache, MySQL, Samba
Cloud	Microsoft Azure

Project Summary

Designed, deployed, and administered a multi-tier enterprise infrastructure environment in Microsoft Azure to simulate real-world IT operations and systems administration. The environment incorporated Active Directory Domain Services, DNS, DHCP, Group Policy, WSUS patch management, SMB file services, IIS web hosting, SQL Server, Linux administration, and backup/recovery operations.

The project focused on centralized identity management, role-based access control, network services, patch management, application hosting, database connectivity, cross-platform administration, and business continuity processes. All services were validated through end-to-end testing including domain authentication, DNS resolution, Group Policy deployment, file share access controls, web application accessibility, SQL connectivity, Linux service validation, and backup verification.

Key Outcomes

- Built and managed an Active Directory domain environment using Windows Server 2022.
- Implemented DNS, DHCP, and Group Policy for centralized administration.
- Configured WSUS for enterprise patch management and update deployment.
- Implemented SMB file services with NTFS permissions and role-based access control.
- Deployed IIS and SQL Server Express in a multi-tier application architecture.
- Configured and administered Ubuntu Linux services including Apache, MySQL, Samba, and OpenSSH.
- Performed backup and recovery validation using Windows Server Backup.
- Demonstrated troubleshooting, security configuration, and infrastructure validation skills across Windows and Linux platforms.